

NO KINETIC DIFFERENCES DURING VARIATIONS OF THE POWER CLEAN IN INEXPERIENCED FEMALE COLLEGIATE ATHLETES

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ABSTRACT

Comfort, P, McMahon, JJ, and Fletcher, C. No kinetic differences during variations of the power clean in inexperienced female collegiate athletes. *J Strength Cond Res* 27(2): 363–368, 2013—Previous research has identified that the second pull phase of the clean generates the greatest power output and that the midhigh variations of the power clean also result in the greatest force and power output in male athletes; however, no research has compared the kinetics of the variations of the power clean in women. The aim of this investigation was to identify any differences between variations of the clean, across a range of loads, in inexperienced female collegiate athletes. Sixteen healthy female collegiate athletes (age 19 ± 2.3 years; height 166.5 ± 3.22 cm; body mass 62.25 ± 4.52 kg; 1 repetition maximum [1RM] power clean 51.5 ± 2.65 kg) performed 3 repetitions of 3 variations (power clean, hang power clean, midhigh power clean) of the power clean at 60, 70, and 80% of their predetermined 1RM power clean, in a randomized and counterbalanced order. A 2-way analysis of variance (3×3 ; load $\times$ variation) revealed no significant differences ($p > 0.05$) in peak power, peak vertical force (F_z) or rate of force development (RFD) between loads or variations of the power clean. There appears to be no advantage in terms of peak power, F_z , or RFD between variations of the clean, in inexperienced female athletes, it is suggested, therefore, that inexperienced athletes intermittently perform different variations of the clean to ensure all round development and technical competence in each variation of the exercise.

KEY WORDS peak power, peak force, rate of force development

INTRODUCTION

Variations of the clean and power clean, incorporating different starting positions (from the floor, hang, and midhigh), are commonly incorporated into strength and conditioning programs. It has

been suggested that such exercises increase an athlete's performance by imitating sport-specific movements, while concurrently using explosive power (20–22), with performance in the hang power clean being correlated to both 20-m sprint performance and countermovement jump performance (14).

The majority of research regarding the kinetic characteristics during performances of the power clean and its variations has focused on the load that achieves peak power output (5,7,16–18). Kawamori et al. (17) found that peak power output during the hang power clean is achieved using a load of 70% of 1 repetition maximum (1RM) power clean. More recently, however, Kilduff et al. (18) found that peak power output during the hang power clean was not significantly ($p > 0.05$) different between loads of 50, 60, 70, 80, or 90% of 1RM power clean. Alternatively, during the power clean, Cormie et al. (5,7) demonstrated that peak power is achieved at a load of 80% 1RM.

Research on experienced weightlifters' technique during the clean has shown that athletes generate peak vertical ground reaction forces (GRF, F_z) during the second pull phase compared with the first pull phase (8). Häkkinen et al. (12) and Souza et al. (19) found similar results, with the second pull displaying the greatest peak F_z at 150% of the system load. Garhammer (9–11) also identified the second pull phase as eliciting the highest power output compared with the first pull in Olympic weightlifters.

Peak power output during the midhigh clean pull has been shown to be achieved at 60% of 1RM (power clean), although the shortest time to peak rate of force development (RFD) was achieved at the lightest load tested (30% of 1RM) (17). Interestingly, time to peak RFD during midhigh clean pulls, at all loads (30, 60, 90, 120% of 1RM power clean) was shorter than time to peak RFD in both countermovement jumps and squat jumps. More recently, Comfort et al. (4) found a significantly greater peak F_z , RFD, and peak power output during the midhigh power clean and the midhigh clean pull compared with both the hang power clean and the power clean, performed at 60% 1RM power clean, respectively.

There is limited evidence to indicate as to which technique variations may be optimal in terms of generating peak power, F_z , and RFD (3,4,8,12,19). The majority of available evidence

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Figure 1. Power clean start position.

has determined that in well-trained male athletes the second pull phase of the power clean results in the greatest F_z (3,4,8–11,19), with only 2 studies identifying that the midhigh power clean and midhigh pull result in a significantly higher peak power, F_z , and RFD compared with other variations of the power clean when performed at 60% 1RM power clean (3,4). Because 60% 1RM power clean has been shown to result in peak power output during the midhigh clean pull (17), this may have biased these findings. The aim of this study was to compare kinetic variable (peak power, peak F_z , and peak RFD)



Figure 2. Hang power clean start position.



Figure 3. Midhigh power clean start position.

during the power clean, hang power clean, and midhigh power clean and to identify any differences, or interactions across loads (60, 70, and 80% 1RM) in inexperienced female collegiate athletes. It was hypothesized that the midhigh power clean would result in higher peak power output because it requires less technical excellence and it has previously been shown to elicit higher peak power, F_z , and instantaneous RFD when compared with the power clean and hang power clean (3,4). In addition, it was also hypothesized that peak power in the midhigh power clean would occur at 60% 1RM power clean in line with previous research (17).

METHODS

Experimental Approach to the Problem

This study employed a within-subjects design, with repeated measures, whereby the dependent variables (peak F_z , peak power, and RFD) were determined while the athletes performed all exercise variations (power clean, hang power clean, and midhigh power clean), at each load (60, 70, and 80% 1RM power clean) while standing on a force platform. These loads were selected because prior research has demonstrated that peak power output is achieved in the midhigh clean pull, hang power clean, and power clean at loads of 60, 70, and 80% of 1RM power cleans, respectively (2–4,7,16–18).

Subjects

Sixteen healthy, but inexperienced (6–12 months performing variations of power cleans regularly) female collegiate athletes (age 19 ± 2.3 years; height 166.5 ± 3.22 cm; body mass 62.25 ± 4.52 kg; 1RM power clean 51.5 ± 2.65 kg)

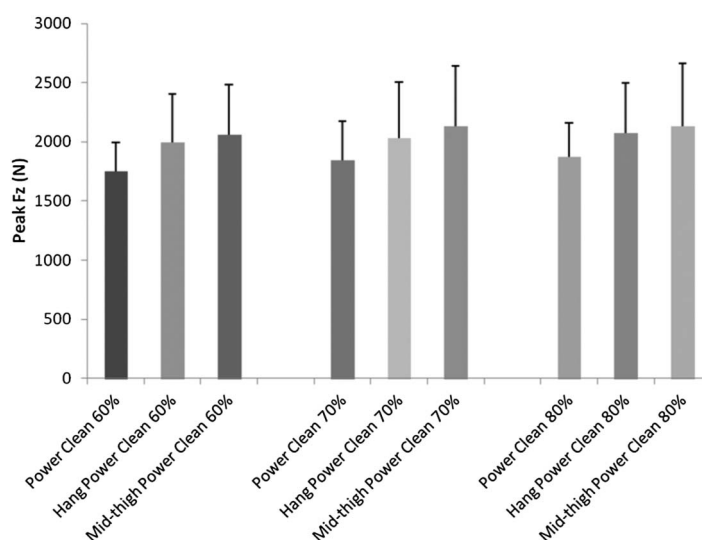


Figure 4. Comparison of peak F_z between variations of the power clean.

participated in this study. The investigation was approved by the Institutional Ethics Review Board, and all the subjects provided written informed consent before participation. The study conformed to the principles of the World Medical Association's Declaration of Helsinki. The participants had previously conducted technique sessions, supervised by a certified strength and conditioning coach to allow familiarization with the protocols and ensure appropriate and consistent technique. All the testing was conducted during the competitive season, at the end of a power mesocycle.

Testing

Before testing, all the subjects performed a standardized dynamic warm up, including each variation of the power clean (4 repetitions, 3 sets) using a standardized load (30 kg)

(Werksan Olympic bar and weights, Moorestown, NJ, USA). The participants were then randomly assigned to perform 1 cluster set of 3 repetitions (30-second rest between repetitions to minimize fatigue) of each exercise in a randomized balanced order. The subjects started with either the power clean (bar starting midway up the shin [Figure 1] and caught in a shallow squat), hang power clean (bar starting in line with the top of the patella [Figure 2] and caught in a shallow squat), and midhigh power clean (bar starting in line with the middle of the thigh [Figure 3] and caught in a shallow squat), in line with previous research (3,4). Countermovement was not permitted during the execution of any lift. Three repetitions of each lift were performed at 60, 70, and 80% of each individual's previously determined 1RM power clean (3 repetitions, of 3 variations of the power clean at 3 loads).

The 1RM power cleans were assessed on 2 separate occasions, 3–5 days apart, to determine reliability following a standardized protocol (1). The heaviest load lifted across the 2 testing sessions was used to calculate the testing loads.

All the lifts were performed with the subjects standing on a force plate (Kistler, Amherst, NY, USA, Model 9286AA, SN 1209740), sampling at 1,000 Hz, interfaced with a laptop. Data were later analyzed using Bioware (Version 3.22; Kistler) to determine peak F_z . Instantaneous RFD was determined by dividing the difference in consecutive F_z readings by the time interval (0.001 seconds) between readings (3,4). Data were

TABLE 1. Comparison of kinetic variables across loads and clean variations.

Load	Midhigh power clean	Hang power clean	Power clean
60% 1RM F_z (N)	2,058.8 ± 423.6	1,994.5 ± 412.7	1,754.2 ± 243.3
RFD ($N \cdot s^{-1}$)	10,966.9 ± 3,559.3	9,034.0 ± 2,946.8	9,475.1 ± 3,713.5
Power (W)	2,695.2 ± 426.7	2,525.9 ± 626.7	2,649.4 ± 610.5
70% 1RM F_z (N)	2,133.9 ± 508.2	2,032.2 ± 470.5	1,846.1 ± 325.3
RFD ($N \cdot s^{-1}$)	10,542.0 ± 3,755.2	10,041.0 ± 3,675.7	9,850.2 ± 2,946.8
Power (W)	2,834.7 ± 626.0	2,702.0 ± 628.4	2,752.0 ± 623.9
80% 1RM F_z (N)	2,131.3 ± 531.9	2,076.3 ± 424.2	1,875.0 ± 285.2
RFD ($N \cdot s^{-1}$)	9,457.5 ± 4,428.9	10,820.9 ± 4,842.8	11,352 ± 4,486.5
Power (W)	2,467.3 ± 640.6	2,475.6 ± 534.4	2,969.9 ± 518.7

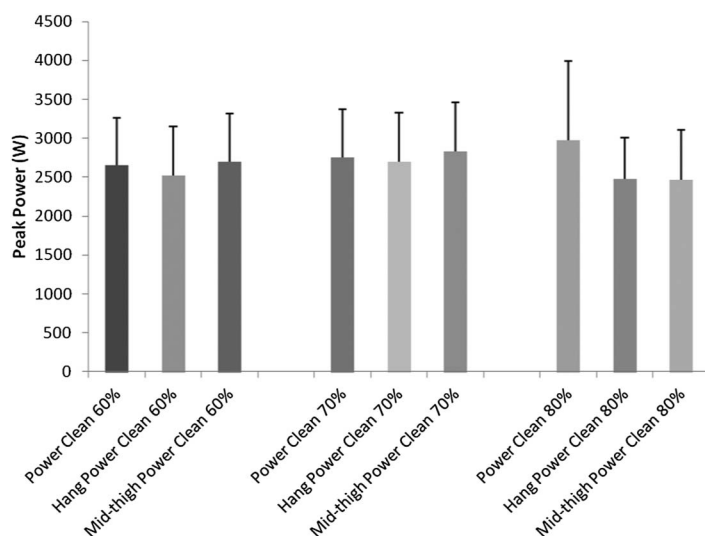


Figure 5. Comparison of peak rate of force development (RFD) between variations of the power clean.

smoothed using a moving average window of 400 milliseconds, in line with previous research (3,4). Velocity of the center of gravity (COG) of the system (barbell + body) was calculated from GRF time data based on the relationship between impulse and momentum in which impulse is equal to the changes in momentum (forward dynamics approach). Power applied to the system was calculated as the product of

velocity of the COG of the system and GRF at each time point (6,13,15). To calculate power in this way, it was important that the initial F_z represented system load (athlete's body mass plus load lifted); consequently, the bar was held slightly above ground level before the onset of the full power clean (Figure 1) (3,4,13,15). The best performance for each variation of the clean under each loading condition was used for further analysis.

Statistical Analyses

A 2 way analysis of variance (ANOVA; load $\times$ exercise variation) with Bonferroni post hoc analysis were conducted to determine if there were any significant differences in the dependent variables (peak F_z , peak RFD, and peak power output) between variations of the power clean, within and between each load. Intraclass correlation coefficients (ICCs) were calculated to determine reliability between 1RM power cleans and to establish reproducibility between repetitions during each exercise variation at each load. Statistical power of the dependent variables was calculated between 0.82 and 0.98 for each variation of the power clean, across loads. An apriori alpha level was set to $p \leq 0.05$. All statistical analyses were conducted using SPSS (version 17.0).

RESULTS

The ICCs demonstrated a high and statistically significant level of reliability for the 1RM power clean ($r = 0.97, p < 0.001$) and a high level of reproducibility between repetitions for the power clean ($r = 0.89, p < 0.01$), hang power clean ($r = 0.91, p < 0.001$) and midthigh power clean ($r = 0.94, p < 0.001$).

Two-way analysis of variance demonstrated no significant differences ($p > 0.05$) in peak F_z between the power clean ($1,754.2 \pm 243.3$ N; $1,846.1 \pm 325.3$ N; $1,875.0 \pm 285.2$ N),

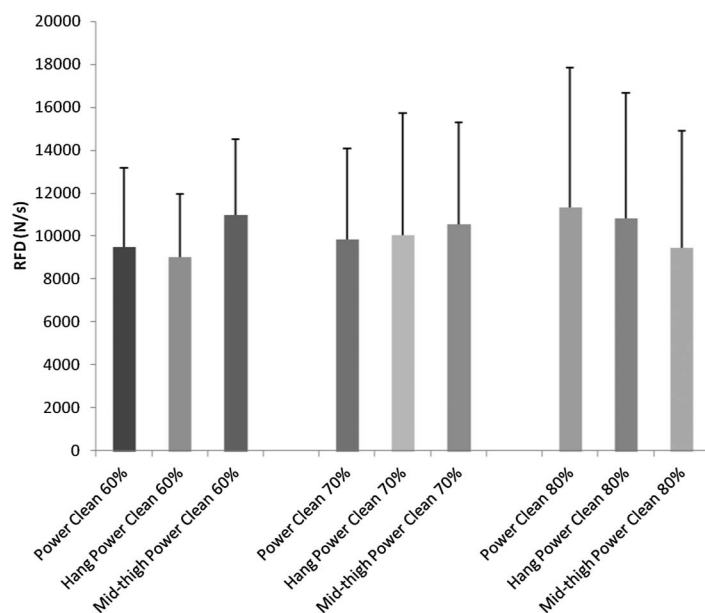


Figure 6. Comparison of peak power output during variations of the power clean.

hang power clean ($1,994.5 \pm 412.7$ N; $2,032.2 \pm 470.5$ N; $2,076.3 \pm 424.2$ N), and the midhigh power clean ($2,058.8 \pm 423.6$ N; $2,133.9 \pm 508.2$ N; $2,131.3 \pm 531.9$ N) performed at 60, 70, and 80% 1RM power clean respectively (Figure 4). Although F_z progressively increased with load, there was no significant difference ($p > 0.05$) in peak F_z across loads, or load by variation interaction (Table 1).

No significant differences ($p > 0.05$) in instantaneous RFD were found between the power clean ($9,475.1 \pm 3,713.5$ N·s⁻¹; $9,850.2 \pm 2,946.8$ N·s⁻¹; $11,352 \pm 4,486.5$ N·s⁻¹), hang power clean ($9,034.0 \pm 2,946.8$ N·s⁻¹; $10,041 \pm 3,675.7$ N; $10,820.9 \pm 4,842.8$ N·s⁻¹), and midhigh power clean ($10,966.9 \pm 3,559.3$ N·s⁻¹; $10,542 \pm 3,755.2$ N·s⁻¹; $9,457.5 \pm 4,428.9$ N·s⁻¹) performed at 60, 70, and 80% 1RM power clean, respectively (Figure 5). There were also no significant differences ($p > 0.05$) in peak power across loads or load by variation interaction (Table 1).

No significant differences ($p > 0.05$) in peak power output were found between the power clean ($2,649.4 \pm 610.5$ W; $2,752.0 \pm 623.9$ W; $2,969.9 \pm 518.7$ W), hang power clean ($2,525.9 \pm 626.7$ W; $2,702.0 \pm 628.4$ W; $2,475.6 \pm 534.4$ W), and midhigh power clean ($2,695.2 \pm 426.7$ W; $2,834.7 \pm 626.0$ W; $2,467.3 \pm 640.6$ W) performed at 60, 70, and 80% 1RM power clean, respectively (Figure 6). There were also no significant differences ($p > 0.05$) in peak RFD across loads or load by variation interaction (Table 1).

DISCUSSION

The findings of this study should aid strength and conditioning coaches during selection of which variations of the clean to perform during different phases of a periodized training program in inexperienced female athletes. These findings are important because they describe the kinetics of power clean variations in inexperienced athletes and highlight the importance of using a range of loads and exercise variations to ensure full development of inexperienced athletes, such as high school and collegiate athletes. The highest peak power output and RFD were achieved during the power clean at 80% 1RM power clean, although this was not significantly greater than any other load or variation of the clean. In contrast, the greatest F_z was observed during the midhigh power clean performed at 70% 1RM power clean, although again this was not significantly greater than any other load or clean variation.

The midhigh power clean resulted in a greater F_z (17.3, 13.7, and 15.4%) compared with the power clean, at loads of 60, 70, and 80% 1RM, respectively; however, this was not statistically significant. Similarly, the midhigh power clean resulted in a greater F_z (3.0, 5.0, and 2.6%) compared with the hang power clean, at loads of 60, 70, and 80% 1RM, respectively, although this was not statistically significant. There were also no significant differences in peak F_z across loads, or load by variation interaction. This is in contrast to previous findings in well-trained rugby league players, which showed significantly greater peak F_z during the midhigh

power clean and the midhigh clean pull ($2,880.2 \pm 236.2$ N) compared with both the power clean ($2,306.2 \pm 240.5$ N) and the hang power clean ($2,442.9 \pm 293.2$ N) (4). The lower peak F_z in this study is likely a result of a lower system load compared with the male rugby league players in the study by Comfort et al. (4), as a product of lower body mass and bar mass used in this study.

Contrary to the previous study by Comfort et al. (4) that found RFD was significantly greater during the midhigh power clean ($15,049.8 \pm 4,415.7$ N·s⁻¹) and the midhigh clean pull ($15,623.6 \pm 3,114.4$ N·s⁻¹) compared with both the power clean ($8,675.8 \pm 2,746.6$ N·s⁻¹) and the hang power clean ($10,314.4 \pm 4,238.2$ N·s⁻¹), this study observed no significant differences in instantaneous RFD during the power clean, hang power clean, and midhigh power clean performed at 60, 70, and 80% 1RM power clean, respectively. There were also no significant differences in peak RFD across loads or load by variation interaction, although the highest RFD was achieved during the power clean performed at 80% 1RM.

In contrast with previous research (4), this study also found no significant differences in peak power output between the power clean hang power clean and midhigh power clean performed at 60, 70, and 80% 1RM power clean, respectively. There were also no significant differences in peak power across loads, or load by variation interaction, although the highest peak power was observed during the power clean performed at 80% 1RM. In line with previous research by Kawamori et al. (16) and Kilduff, et al. (18), this study also found no significant difference in peak power output during the hang power clean at loads of 60, 70, and 80% 1RM power clean. Even though no significant differences were found in the kinetic data across exercise variations or loads, it was observed that the second pull phase of each variation of the exercises generated the greatest peak F_z and peak power output, as noted by previous research (9–11).

The fact that no significant differences were found in terms of peak F_z , peak power and RFD across exercise variations in inexperienced female athletes, in contrast to prior research (4), is likely because of the level of experience and proficiency of the athletes. We acknowledge that only a small range of loads 60–80% was used in this study, which are not as high as the loads sometimes used in training; however, because of the limited experience of these athletes and the fact that optimal loads have previously been identified at $\leq 80\%$ 1RM higher loads are unlikely to elicit greater power output and also more likely to result in failure of the lift. Future research should consider determining if the lack of differences in these kinetic variables changes as experience and peak power output increases.

PRACTICAL APPLICATIONS

The results of this study demonstrate that there is no advantage in terms of peak power, peak F_z , or RFD between variations of the clean, in inexperienced athletes. The

midthigh power clean may offer practical benefits in terms of its less complex nature, whereas the hang power clean and the power clean may aid in the development of the stretch-shorten cycle. Because no variation of the power clean appears to demonstrate a benefit in terms of kinetic measurements, it is suggested that inexperienced athletes intermittently perform different variations of the clean to ensure all round development and technical competence. In addition, loads of 60% 1RM power clean may be advantageous, to allow the athlete to focus on technique, rather than the load on the bar, until they are more experienced in the exercises, as no differences in peak power, $F_{\dot{v}}$, or RFD were noted between 60, 70, and 80% 1RM power clean.

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